Virtual Try-On: main findings

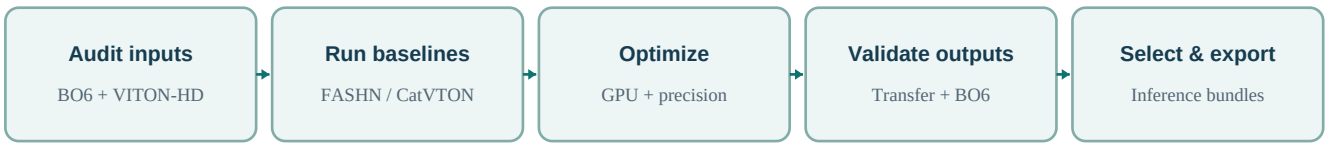
Tutor report | Aymen Dhieb | 8 October 2026

Selected baseline: FASHN v1.5
float16 / 15 steps / guidance 1.5 / seed 42.
Measured median: **34.70 s** on six changed-garment cases and **35.86 s** on eight BO6 catalogue garments, using one Kaggle Tesla T4.

Objective and contribution

Generate a visual simulation of a person wearing a selected BO6 wardrobe garment. The notebook work established pretrained baselines, compared their outputs, diagnosed runtime bottlenecks, tested catalogue inputs and exported inference bundles.

Research pipeline completed



Data preparation: concrete results

Audit / preparation	Result
BO6 image and metadata audit	44,441 images; 44,424 metadata rows
Quality checks	0 corrupt images; 22 malformed metadata records quarantined
Duplicate content	1,431 rows in exact duplicate-content groups
VTON category mapping	15,354 initial candidates: tops, bottoms and one-pieces
VITON-HD preparation	4,000 inventoried person/garment cases; derived binary masks

BO6 supplies catalogue garments; VITON-HD supplies person/garment benchmark cases. These are different roles. The exported candidate set includes one Accessories record; an Apparel-only filter would retain 15,353 candidates.

What has been demonstrated
Working pretrained generation, a controlled development comparison and a BO6 garment demonstration. **No completed training or LoRA run is evidenced.** Web integration and mandatory Hybrid Avatar remain incomplete in the supplied evidence.

Evidence basis: supplied research ZIP and inference bundles. The .ipynb was not supplied for this report; exported logs and images were inspected, without rerunning GPU inference.

Measured results and model choice

Controlled development comparison

The same 12 cases and seed 42 were used: six paired reconstructions and six changed-garment transfers. All **36 main runs produced images**, across three configurations. Image generation success does not establish garment fidelity.

Configuration / set	Output size	Cases	Median (s)
FASHN / 30 steps	576 x 768	12	71.44
CatVTON / 30 steps	384 x 512	12	7.46
CatVTON / 30 steps	768 x 1024	12	73.50
FASHN / 15 steps / transfers	576 x 768	6	34.70
FASHN / 15 steps / BO6	576 x 768	8	35.86

Why FASHN was selected

Simpler person-input preparation: the selected configuration needs no externally supplied person mask. The recorded single-reviewer comparison found quality competitive with higher-resolution CatVTON on changed garments. This supports a practical selection; it does not establish a universal quality winner.

Runtime improvements

Change	Measured evidence / interpretation
BF16 to float16	447.91 s to 65.49 s on one diagnostic sample: 6.84x latency reduction. T4 has no native BF16 support.
Sampling steps	15-step transfer median: 34.70 s; BO6 median: 35.86 s. Visual comparisons are shown on the next pages.
GPU preprocessing	Recorded FASHN detector and pose sessions advertise CUDAExecutionProvider; CPU fallback was addressed.

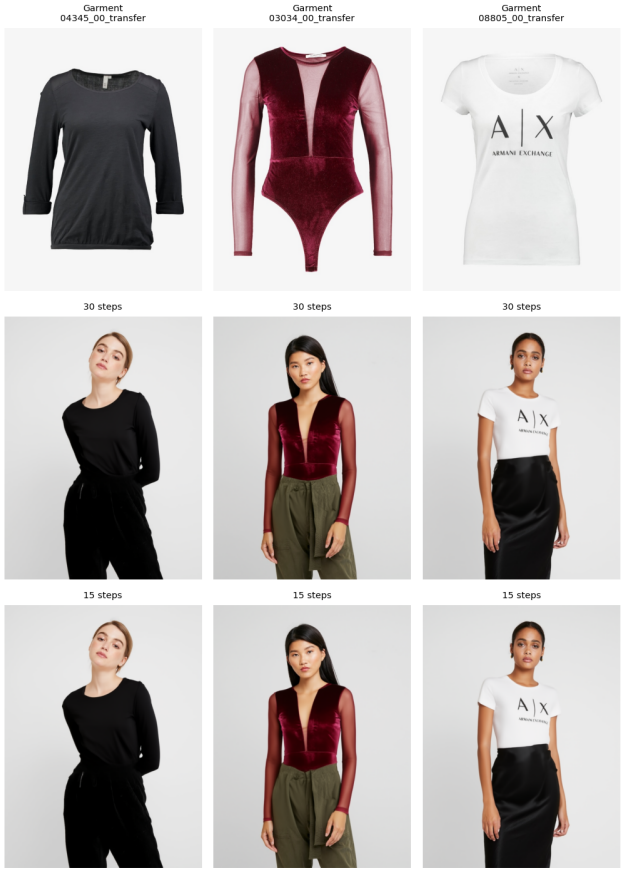
How to read these numbers
Resolutions differ, and the final 15-step sets differ from the 12-case baseline. Latency covers inference, **not the full website journey**; initialization is logged separately. CatVTON timings exclude mask creation. PyTorch memory measurements exclude some ONNX/driver overhead.

No completed numeric quality ratings or broad held-out evaluation are evidenced. Paired reconstruction may favour a maskless model because its input already shows the target garment. Changed-garment examples are more representative of the intended use.

Transfer examples: colour, shape and logo

Main idea: fewer steps can retain a useful result
These three saved cases compare the same target garment at 30 and 15 steps. They support the faster setting visually on these examples; they do not prove equivalence for all garments.

Read each column from top to bottom: target garment, 30-step output, 15-step output. The original person photo is not included in this exported comparison.



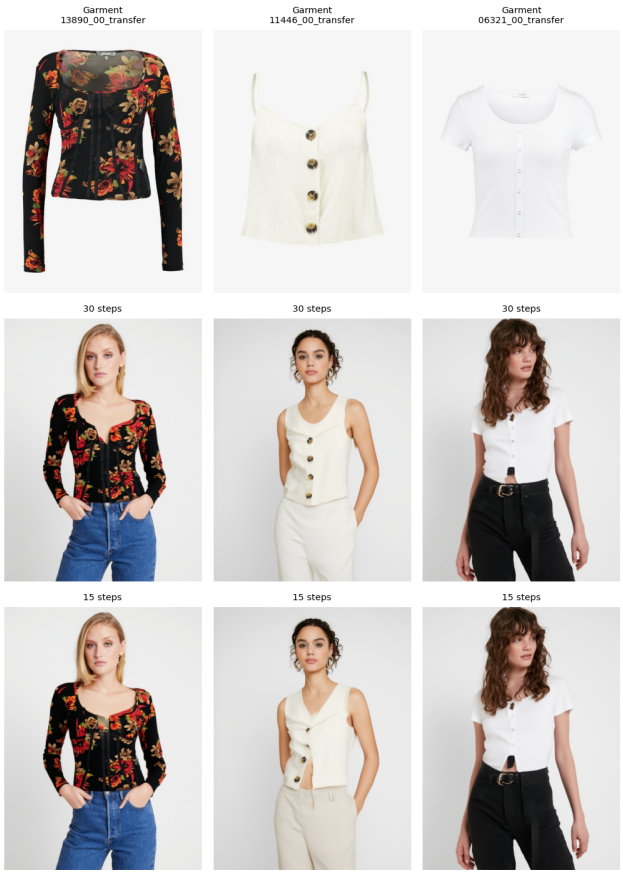
Example	Observed result
Dark long-sleeve	Long sleeves and broad silhouette transfer, but the output looks darker than the catalogue input.
Burgundy bodysuit	Burgundy colour and central panel are recognizable; the lower portion is hidden by the trousers.
White logo T-shirt	Main lettering is recognizable; small printed text and exact layout should not be assumed faithful.

Figure 1. First three columns of the exported FASHN 30-versus-15-step transfer sheet. Qualitative observations from saved images; no numerical fidelity score.

Transfer examples: patterns and failure analysis

Main idea: faster inference does not remove model failures
The cream camisole fails at both sampling settings. A usable generation must preserve the target garment and avoid changing unrelated clothing.

Rows: target garment / 30 steps / 15 steps.



Example	Observed result
Floral long-sleeve	Floral appearance and long sleeves remain visible; exact pattern placement changes.
Cream camisole - failure	Thin straps become a wider, waistcoat-like shoulder shape; unrelated trousers also become cream.
White buttoned top	White colour, buttons and short sleeves transfer; hem and front opening differ across settings.

Figure 2. Remaining three transfer cases. These examples illustrate shape, pattern and unrelated-clothing errors. The prior informal review judged five of six outputs broadly similar across step settings.

BO6 wardrobe examples: four tops

Main idea: BO6 catalogue images can drive VTON
The same benchmark person was used with eight BO6 garments. Both flat-lay and worn catalogue photos were tested. The four examples below show tops at the selected 15-step setting.



Original person photo
White top and black trousers. Compare these with the generated images to identify garment changes and unintended edits.

Top row: BO6 catalogue garment. **Bottom row:** generated result; labels show rounded inference time.



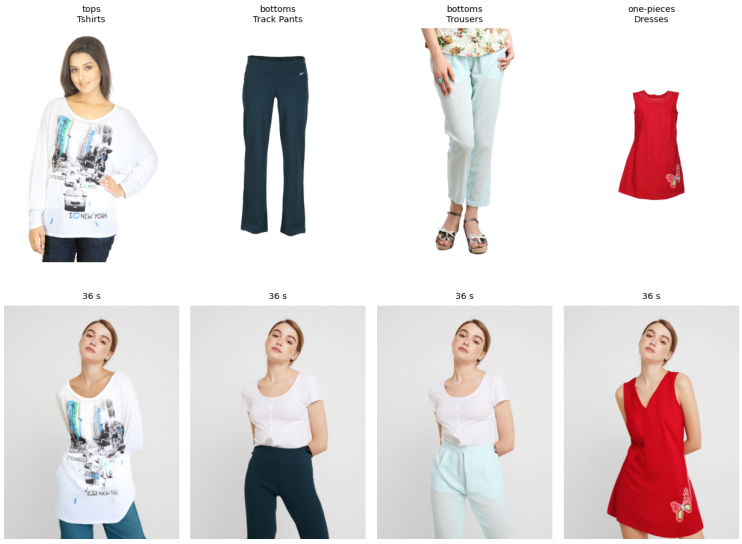
Garment	Observation
Orange T-shirt	Orange colour, short sleeves and broad print transfer. Correct flat-lay photo-type metadata was used.
Red ruffled top	Ruffles, red colour and central front details remain visible on the new person.
Purple printed T-shirt	Colour and large graphic transfer; fine lettering and graphic proportions remain approximate.
Yellow top	Colour, layered neckline and sleeve style transfer, with changed draping.

Figure 3. Four of eight corrected-photo-type BO6 outputs. These are demonstrations with one person, not a generalization or accuracy estimate.

BO6 examples, limitations and next milestone

Main idea: catalogue compatibility is demonstrated
All eight corrected BO6 runs produced images; median **35.86 s**. A white printed top also transfers unrelated catalogue jeans, showing why output quality must be reviewed.

Same original person as page 5. Top: catalogue input. Bottom: generated result.



Garment	Observation
White printed top - failure	Printed appearance transfers, but the original black trousers change to blue jeans from the catalogue photo.
Track pants	Dark colour, trouser silhouette and small logo transfer; white upper garment broadly remains.
Light trousers	Light colour and loose trouser shape transfer from a worn catalogue image.
Red dress	One-piece transfer demonstrated; neckline and overall silhouette differ from the input.

What the tutor should retain

Contribution: data audit, pretrained comparison, measured runtime optimization, visual failure analysis and exported workers. **Limits:** small development set, informal visual assessment, one person for BO6 tests, no LoRA/training result, and no complete application validation.

Next milestone: repair FASHN checksum checks that still reference temporary download-cache files; verify a fresh bundle restore and real inference; then connect the worker to the application and test more people. Mandatory Hybrid Avatar requires automatic initialization plus manual correction, save/reload and intended product use.

Evidence references: bo7_research_evidence (1).zip: audit summary, model_selection.json, runs/*.jsonl and the two contact sheets reproduced here. fashn_inference_bundle (1).zip and catvton_base_inference_bundle (1).zip: configuration and exported inference workers. All illustrated images are existing exported results; no new inference was performed.